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Hot Chips Day 1: Key Themes and Takeaways

CITI'S TAKE

We attended Hot Chips 2026 Day 1 in Palo Alto, CA yesterday. Overall, the key message was that the industry is moving from a GPU-centric AI buildout toward an infrastructure-centric AI buildout, where CPUs, networking, memory, packaging, and system integration all become increasingly critical alongside accelerators. We summarize key themes and takeaways below:

1. CPUs Are Becoming the "Orchestration Layer" for Agentic AI

The most consistent theme across Intel, AMD, Nvidia, Fujitsu, and Arm was that CPU role is evolving toward orchestration, data movement, scheduling, and agent coordination. AI is creating a CPU revival narrative. The winning CPUs are not necessarily those with the highest raw compute, but those that can efficiently manage massive AI workflows and data movement.

- Nvidia repeatedly emphasized that agentic AI is a workflow problem, not just a compute problem. Vera CPU is explicitly designed to maximize agent performance per core, data processing, and I/O.
- Fujitsu highlighted that agentic AI increases CPU demand for orchestration and peripheral processing.
- Intel Wildcat Lake was designed to handle AI orchestration through its NPU and CPU cores.

2. Power Efficiency Is Becoming the Primary Design Constraint

Virtually every compute vendor focused on performance-per-watt rather than absolute performance. The industry is transitioning from maximizing FLOPS to maximizing performance under power constraints. This favors advanced process nodes, packaging innovation, and system architecture improvements. Examples highlighted included:

- Intel Wildcat Lake reduced die sizes, memory requirements, power consumption, and packaging costs.
- Nvidia Vera emphasized operation in power-constrained data centers.
- Fujitsu Monaka positioned itself as a "green AI datacenter" processor.
- Intel Diamond Rapids highlighted balancing performance, scalability, security, and efficiency.

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- AMD's long-term goal is a 20x improvement in rack-scale efficiency by 2030.

3. Packaging and System Architecture Matter as Much as Process Technology

Future differentiation will increasingly come from packaging, chiplets, interconnects, and system integration rather than from process node leadership alone. Examples highlighted included:

- Intel Wildcat Lake which eliminated the base die, reduced compute area by ~30%, moved to MCP packaging, and uses UCIe for die-to-die communication.
- Intel Diamond Rapids uses Fan-out Fabric architecture, modular compute blocks, and unified memory and I/O fabrics.
- AMD MI400 uses CoWoS-L packaging, multi-chiplet design, separate accelerator, cache/fabric, and I/O dies.
- Nvidia put major focus on system integration across CPU, GPU, network, and storage.

4. Memory and Data Movement Are Emerging as the Bottlenecks

AI system performance is increasingly limited by memory bandwidth, interconnects, and data movement rather than pure compute. Examples highlighted include:

- Nvidia Rubin GPU has scalable coherency architecture, massive I/O, NVLink 6 with 3x lower latency and 10x higher packet rate versus Ethernet. Memory capacity is a tradeoff that Nvidia makes for every generation.
- AMD MI455X has 432GB HBM4, 23.3 TB/s bandwidth, significant increases in memory and interconnect performance.
- Intel Diamond Rapids has 16-channel memory architecture and unified memory fabric.

5. Custom Silicon Continues to Proliferate

Across the conference there was a recurring theme that highly specialized workloads increasingly justify custom silicon.

AI is creating more opportunities for application-specific processors and accelerators, benefiting foundries, packaging suppliers, and custom silicon ecosystems. Waymo provided a strong real-world example using an internally developed sensor-fusion processor, built on TSMC 5nm, and purpose-built for autonomous driving workloads.

6. Physical AI Is the Next Major Compute Wave

While today's AI spending is concentrated in datacenters, future AI growth extends into robotics, autonomous vehicles, and "physical AI." Physical AI requires low latency, low power, high reliability, and unique safety compute requirements. Waymo arguably presented the most important long-term demand signal with 200M+ autonomous miles driven, 20M+ paid autonomous trips, multimodal AI (camera, lidar, radar, microphones), compute operating below 1 kW and long-term goal of eventually running a 1 trillion parameter model inside a vehicle.

7. ARM's Share Gains in AI Infrastructure Continue

ARM continues to gain relevance in hyperscale AI infrastructure, particularly where power efficiency is critical. ARM AGI pushes to 136 cores and 100B+ transistors. Arm stated it now has 60%+ share of host CPUs for AI platforms. Moreover, Nvidia Vera uses custom ARM-based Olympus cores and Fujitsu Monaka is also ARM-based.

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